Joseph Dhliwayo

Hamden, Connecticut, United States · Fluent English

Cybersecurity Engineer

mrdhliwayojoseph@gmail.com
codewithjoe.online
linkedin.com/in/joseph-dhliwayo-12b90b17b

Cybersecurity engineer completing an MSc in Cybersecurity at Quinnipiac University in December 2026, with a capstone, AURA, that uses machine learning and AI to detect compromised and abusive accounts in real time. Backed by AWS and Cisco security training and hands-on experience securing authentication, payments, APIs and databases for live platforms, including Pekugara, a housing app on the App Store and Google Play.

EDUCATION & CERTIFICATIONS

Master of Science in Cybersecurity

Quinnipiac University · Completing December 2026 · Hamden, Connecticut · Capstone: AURA, AI-driven user risk analytics

Bachelor of Science Honours in Information Systems

Completed · Zimbabwe

AWS Cloud Security

In progress · Amazon Web Services

AWS Academy: Cloud Architecting

Amazon Web Services · Issued May 2026

AWS Academy: Cloud Foundations

Amazon Web Services · Issued Dec 2025

AWS Academy: Cloud Security Foundations

Amazon Web Services · Issued Dec 2025 · Verify on Credly

Cisco: Cyber Threat Management · Network Support & Security · Intro to Cybersecurity · Networking Basics

Cisco Networking Academy · Issued May–Jul 2024

SECURITY SKILLS

Threat detection

Anomaly detection (Isolation Forest, scikit-learn), detection rules, alert correlation, real-time alerting

Application security

Secure coding, input validation, output encoding, SQL injection, XSS and CSRF prevention

Authentication & sessions

Password hashing (bcrypt, scrypt), signed tokens, httpOnly and SameSite cookies

Cryptography

AES-256 encryption, HMAC-SHA256 signing, timing-safe comparison, TLS troubleshooting

Access control & config review

PostgreSQL row-level security, privileged function review, webhook signature checks

Cloud, privacy & AI

AWS, secrets management, HMAC pseudonymisation, data minimisation, verified LLM output

EXPERIENCE

Capstone Project: AURA, AI-Driven User Risk Analytics

2026

Quinnipiac University · MS Cybersecurity · Piloted on Pekugara

- Designed and built a system that detects compromised and abusive accounts by combining Isolation Forest anomaly detection with detection rules
- The hybrid detector reached 0.87 F1 at a 0.09% false-positive rate across 10 test runs, about a fifth of the false alarms of rules alone
- Built read-only database configuration checks that found 18 of 20 planted weaknesses with no false positives, and two real issues in a production codebase

- Real-time alerts reach administrators by WhatsApp, SMS or email in a median 39 ms, with VPN, proxy and Tor detection
- AI explains each alert from pseudonymised, minimised data; explanations not backed by evidence are rejected, and admins approve every action

Founder & Lead Developer

2024 – Present

Pekugara · Hamden, United States

- Built and secured a housing and rental platform used by renters and landlords across Zimbabwe, live on the App Store and Google Play
- Secured mobile-money payments (Pesepay / EcoCash) with AES-256-CBC encryption and diagnosed TLS-level issues in a non-standard payment API
- Wrote PostgreSQL row-level security policies so each user can only reach their own data, and built admin tools to vet landlords and remove fraudulent accounts

Freelance Web Developer

2022 – Present

Global Website Solutions · Remote

- Built websites and business systems for clients in the US and Zimbabwe; implemented login systems with salted password hashing (bcrypt, scrypt), HMAC-signed sessions checked with timing-safe comparison, and httpOnly, SameSite cookies
- Prevented injection and cross-site scripting with parameterized SQL, Zod validation and output escaping, and verified HMAC-SHA256 webhook signatures so forged WhatsApp and EcoCash requests are rejected
- Built role-based admin access and revocable, time-limited signed invite links for a live radio studio, with secrets kept out of code

SELECTED PROJECTS

On Point Meta Media

onpointmetamedia.com

Live radio station with a password-protected studio, signed sessions, and HMAC invite links that can be revoked at once.

WhatsApp Remit Bot

Private build

Payment chatbot that verifies HMAC-SHA256 signatures on WhatsApp and EcoCash webhooks before acting on them.

Gata Gadgets

gatagadgets.store

Point-of-sale and inventory system with bcrypt-hashed logins, separate staff and admin roles, and escaped output.

The Frontline

thefrontlinezw.com

News site with an editor dashboard secured by scrypt password hashing, signed httpOnly cookies and input validation.

TOOLS & PLATFORMS

Python · FastAPI · scikit-learn · pandas · Claude API · Node.js · TypeScript · Next.js · PostgreSQL · Supabase · Neon · AWS · Vercel · Git & GitHub · Twilio

REFERENCES

Available upon request.